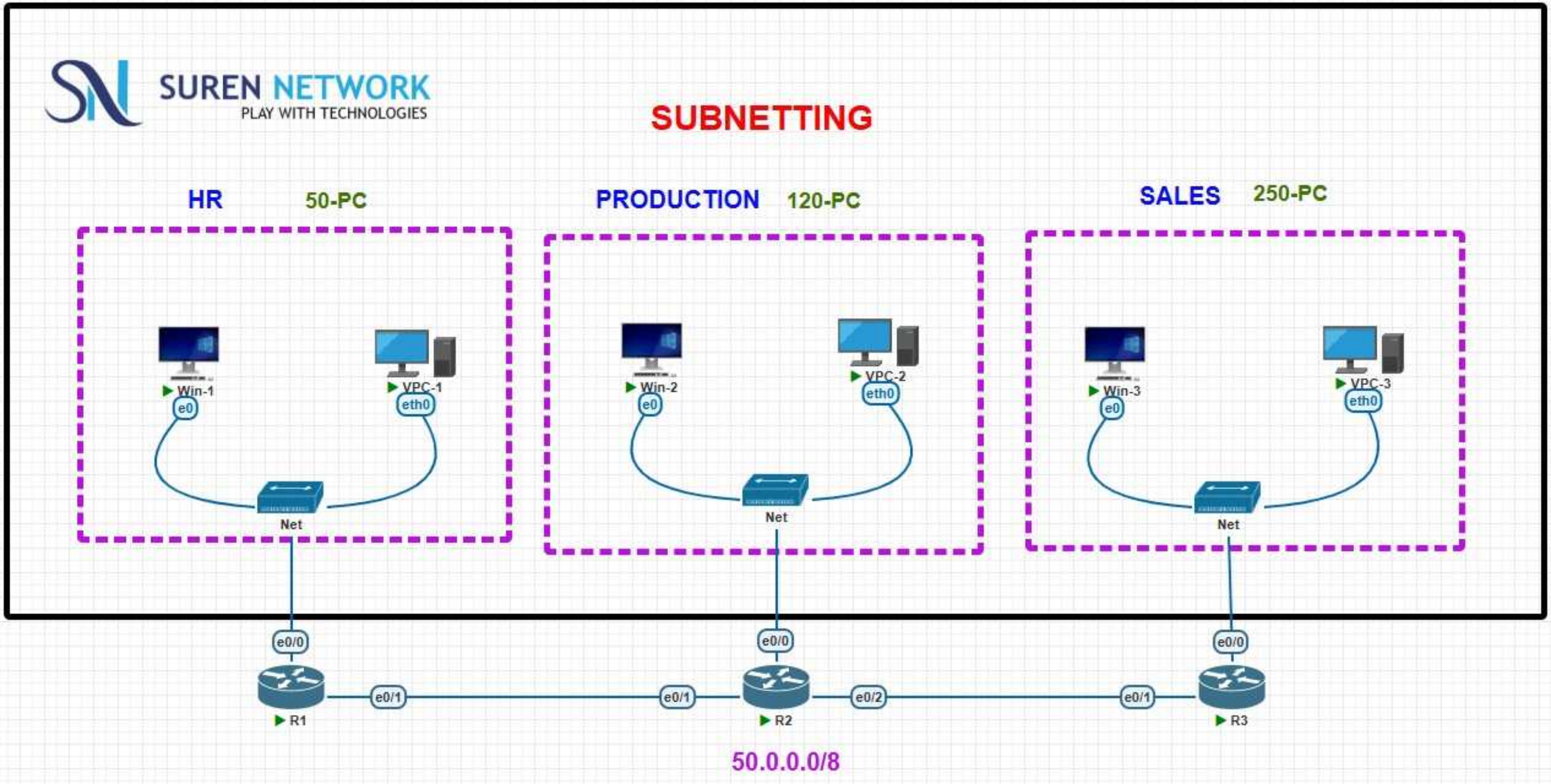

SUREN NETWORK
PLAY WITH TECHNOLOGIES
SUBNETTING
HR
50-PC
PRODUCTION
120-PC
SALES
250-PC
Win-1
e0
VPC-1
eth0
Net
Win-2
e0
VPC-2
eth0
Net
Win-3
e0
VPC-3
eth0
Net
e0/0
e0/1
R1
e0/0
e0/1
e0/2
R2
e0/0
e0/1
R3
50.0.0.0/8

C:\WINDOWS\system32\cmd.exe

```
Ethernet adapter Local Area Connection:

        Connection-specific DNS Suffix  . :
        IP Address. . . . . . . . . . . . : 50.0.1.130
        Subnet Mask . . . . . . . . . . . : 255.255.255.192
        Default Gateway . . . . . . . . . : 50.0.1.129

C:\Documents and Settings\user>ping 50.0.1.1

Pinging 50.0.1.1 with 32 bytes of data:

Reply from 50.0.1.1: bytes=32 time=1ms TTL=254
Reply from 50.0.1.1: bytes=32 time=1ms TTL=254
Reply from 50.0.1.1: bytes=32 time<1ms TTL=254
Reply from 50.0.1.1: bytes=32 time=1ms TTL=254

Ping statistics for 50.0.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Documents and Settings\user>ping 50.0.1.127

Pinging 50.0.1.127 with 32 bytes of data:

Reply from 50.0.1.194: bytes=32 time<1ms TTL=254
Reply from 50.0.1.194: bytes=32 time=1ms TTL=254
Reply from 50.0.1.194: bytes=32 time<1ms TTL=254
Reply from 50.0.1.194: bytes=32 time=1ms TTL=254

Ping statistics for 50.0.1.127:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Documents and Settings\user>ping 50.0.1.130

Pinging 50.0.1.130 with 32 bytes of data:

Reply from 50.0.1.130: bytes=32 time<1ms TTL=128
Reply from 50.0.1.130: bytes=32 time<1ms TTL=128
Reply from 50.0.1.130: bytes=32 time<1ms TTL=128
Reply from 50.0.1.130: bytes=32 time<1ms TTL=128

Ping statistics for 50.0.1.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>ping 50.0.1.129

Pinging 50.0.1.129 with 32 bytes of data:

Reply from 50.0.1.129: bytes=32 time<1ms TTL=255
Reply from 50.0.1.129: bytes=32 time=1ms TTL=255
Reply from 50.0.1.129: bytes=32 time<1ms TTL=255
Reply from 50.0.1.129: bytes=32 time=1ms TTL=255
```

Monday, October 12, 2026
Start
C:\WINDOWS\system...
4:57 PM

QEMU (Win-2)

C:\WINDOWS\system32\cmd.exe

```
Ethernet adapter Local Area Connection:

        Connection-specific DNS Suffix  . :
        IP Address. . . . . . . . . . . . : 50.0.1.2
        Subnet Mask . . . . . . . . . . . : 255.255.255.128
        Default Gateway . . . . . . . . . : 50.0.0.1

C:\Documents and Settings\user>ping 50.0.1.130

Pinging 50.0.1.130 with 32 bytes of data:

Reply from 50.0.1.130: bytes=32 time=3ms TTL=126
Reply from 50.0.1.130: bytes=32 time=1ms TTL=126
Reply from 50.0.1.130: bytes=32 time=1ms TTL=126
Reply from 50.0.1.130: bytes=32 time=1ms TTL=126

Ping statistics for 50.0.1.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 1ms

C:\Documents and Settings\user>ping 50.0.0.2

Pinging 50.0.0.2 with 32 bytes of data:

Reply from 50.0.0.2: bytes=32 time=2ms TTL=126
Reply from 50.0.0.2: bytes=32 time=1ms TTL=126
Reply from 50.0.0.2: bytes=32 time=1ms TTL=126
Reply from 50.0.0.2: bytes=32 time=1ms TTL=126

Ping statistics for 50.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>ping 50.0.0.198

Pinging 50.0.0.198 with 32 bytes of data:

Control-C
^C
C:\Documents and Settings\user>ping 50.0.1.198

Pinging 50.0.1.198 with 32 bytes of data:

Reply from 50.0.1.198: bytes=32 time=1ms TTL=254
Reply from 50.0.1.198: bytes=32 time=1ms TTL=254
Reply from 50.0.1.198: bytes=32 time<1ms TTL=254
Reply from 50.0.1.198: bytes=32 time<1ms TTL=254

Ping statistics for 50.0.1.198:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Documents and Settings\user>ping 50.0.1.193

Pinging 50.0.1.193 with 32 bytes of data:

Reply from 50.0.1.193: bytes=32 time=1ms TTL=254
```

Start | C:\WINDOWS\system... | 5:00 PM

QEMU (Win-3)

C:\WINDOWS\system32\cmd.exe

```
        Connection-specific DNS Suffix  . :
        IP Address. . . . . . . . . . . . : 50.0.0.2
        Subnet Mask . . . . . . . . . . . : 255.255.255.0
        Default Gateway . . . . . . . . . : 50.0.0.1

C:\Documents and Settings\user>ping 50.0.1.130

Pinging 50.0.1.130 with 32 bytes of data:

Reply from 50.0.1.130: bytes=32 time=4ms TTL=125
Reply from 50.0.1.130: bytes=32 time=3ms TTL=125
Reply from 50.0.1.130: bytes=32 time=2ms TTL=125
Reply from 50.0.1.130: bytes=32 time=2ms TTL=125

Ping statistics for 50.0.1.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 4ms, Average = 2ms

C:\Documents and Settings\user>ping 50.0.1.193

Pinging 50.0.1.193 with 32 bytes of data:

Reply from 50.0.1.193: bytes=32 time=1ms TTL=253
Reply from 50.0.1.193: bytes=32 time=1ms TTL=253
Reply from 50.0.1.193: bytes=32 time=1ms TTL=253
Reply from 50.0.1.193: bytes=32 time=1ms TTL=253

Ping statistics for 50.0.1.193:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\Documents and Settings\user>ping 50.0.1.2

Pinging 50.0.1.2 with 32 bytes of data:

Reply from 50.0.1.2: bytes=32 time=2ms TTL=126
Reply from 50.0.1.2: bytes=32 time=1ms TTL=126
Reply from 50.0.1.2: bytes=32 time=1ms TTL=126
Reply from 50.0.1.2: bytes=32 time=1ms TTL=126

Ping statistics for 50.0.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>ping 50.0.1.197

Pinging 50.0.1.197 with 32 bytes of data:

Reply from 50.0.1.197: bytes=32 time=1ms TTL=254
Reply from 50.0.1.197: bytes=32 time=1ms TTL=254
Reply from 50.0.1.197: bytes=32 time=1ms TTL=254
Reply from 50.0.1.197: bytes=32 time=1ms TTL=254

Ping statistics for 50.0.1.197:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

C:\WINDOWS\system32\cmd.exe

Start

C:\WINDOWS\system...

5:06 PM